Skills Summary

Solving and Graphing One-Step Inequalities

Use this reference while completing your worksheet or when studying.

Solve it like an equation, with one extra rule. Undo the single operation on both sides. Adding or subtracting never changes the sign; **multiplying or dividing by a negative reverses it** ($<$ becomes $>$, $\leq$ becomes $\geq$).

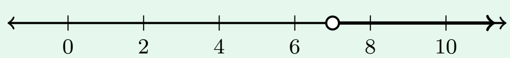
Graph it. Open circle $\circ$ for $<$ or $>$ (boundary not included), solid dot $\bullet$ for $\leq$ or $\geq$ (boundary included). Then shade the ray that holds the numbers that work.

Always test one number from the shaded side back in the original inequality. One test catches a missed flip every time.

1. Undo an addition or a subtraction

If a number is added to the variable, subtract it from both sides; if it is subtracted, add it. The inequality sign never moves for these two operations, even when the answer comes out negative.

Example: Solve and graph $x + 3 > 10$.
Step 1. Only one thing is being done to x — 3 is added — so undo it by subtracting 3, and leave the sign alone.
Step 2. $x + 3 - 3 > 10 - 3$ gives $x > 7$. Strict $>$, so an open circle at 7 shaded right.



Solution: $\boldsymbol{x > 7}$

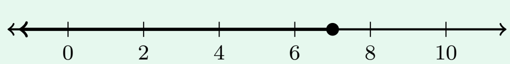
Try it: Solve and graph $k - 4 \leq 1$. Which kind of circle goes at the boundary?

check: $\boldsymbol{k \leq 5}$ (solid dot, shaded left)

2. Undo a multiplication or a division

If the variable is multiplied by a positive number, divide both sides by it; if it is divided, multiply. Positive multiplier or divisor means the sign stays put — a negative number in the *answer* is not a reason to flip.

Example: Solve and graph $5x \leq 35$.
Step 1. The variable is multiplied by 5, so divide both sides by 5; 5 is positive, so the sign does not move.
Step 2. $\frac{5x}{5} \leq \frac{35}{5}$ gives $x \leq 7$. The $\leq$ includes the boundary, so a solid dot at 7 shaded left.



Solution: $\boldsymbol{x \leq 7}$

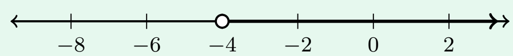
Try it: Solve and graph $\frac{p}{2} > 3$.

check: $p > 6$ (open circle, shaded right)

3. Flip when you multiply or divide by a negative

Multiplying by a negative reflects every number across 0, so the larger one lands on the smaller side: $3 > 1$, but $-3 < -1$. That is why the sign must reverse the moment you divide or multiply by a negative — and only then.

Example: Solve and graph $-3x < 12$.
Step 1. The coefficient is negative, so the step that isolates x is a division by -3 — the one move that reverses the sign.
Step 2. $\frac{-3x}{-3}$ and $\frac{12}{-3}$ give x and -4, and $<$ turns into $>$: $x > -4$. Test $x = 0$: $-3(0) = 0 < 12$ is true, and 0 is in the shaded part.



Solution: $\boldsymbol{x > -4}$

Try it: Solve and graph $-w \geq 5$. (The coefficient is -1.)

check: $w \leq -5$ (solid dot, shaded left)

(!) Watch out: the flip rule keys on what you multiply or divide *by*, never on a minus sign that merely appears in the problem. In $6w \leq -18$ the divisor is 6, so nothing flips; in $-2x > 8$ the divisor is -2, so it does.